



Chapter 3, Capacitance and Inductance

In this chapter, we consider energy storage elements. Similar to resistors these elements are passive; that is, they do not supply power. This chapter describes the relationship between voltage, current, power and energy for inductors and capacitors. In addition, it explains the effect of connecting these components in series and in parallel.

The Capacitor

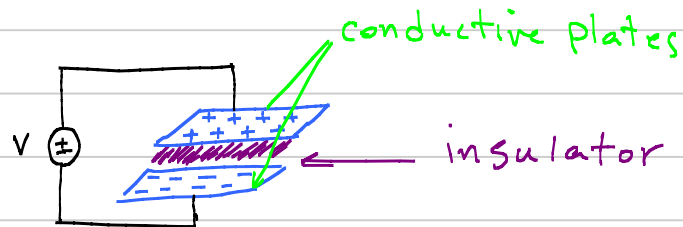
A capacitor is an electrical component that consists of two conductors separated by an insulator. When a voltage is applied between the two conductors, a positive charge is deposited on one conductor and a negative charge on the other. The amount of charge stored in the conductors is proportional to the voltage.

$$q = CV$$

C : capacitance in Farads (F)

V : voltage (V)

q : charge in coulombs



$$1 \text{ Farad} = \frac{1 \text{ Coulomb}}{\text{Volt}}$$

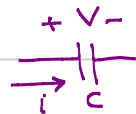
$$C = \frac{\epsilon A}{d}$$

A = Area of each plate

d = distance between the plates

ϵ = permittivity of the dielectric material

$$q = CV$$



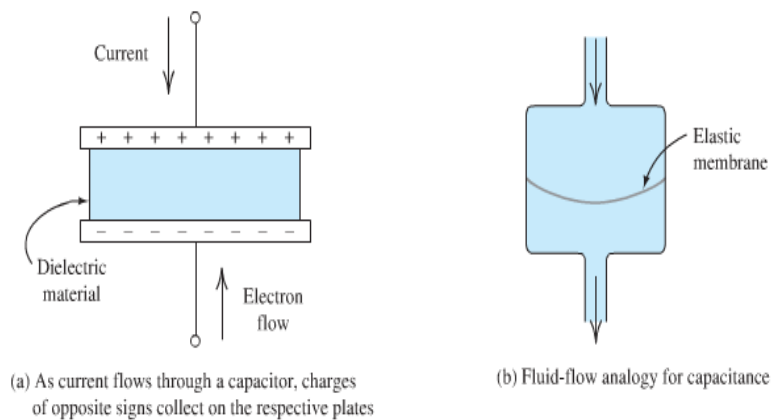
$$\frac{dq}{dt} = C \frac{dV}{dt}$$

$$i = C \frac{dV}{dt}$$

(if i is in the direction of voltage drop)

Fluid flow analogy for a capacitor

In terms of fluid flow analogy, a capacitor represents a reservoir with an elastic membrane separating the inlet and outlet. As the fluid flows into the inlet, the membrane is stretched, creating a force (analogous to capacitor voltage) that opposes further flow. The displaced fluid volume is analogous to the charge stored in the capacitor.



voltage in terms of current

$$i = c \frac{dv}{dt}$$

$$i dt = c dv$$

$$\int_{t_0}^t i dt = \int_{V(t_0)}^{V(t)}$$

$$V(t) = \frac{1}{c} \int_{t_0}^t i dt + V(t_0)$$

(if i is in the direction of voltage drop)

Power and Energy

$$P = vi = c v \frac{dv}{dt}$$

$$W_{(t)} = \int_{t_0}^t P dt = c \int_{t_0}^t v \frac{dv}{dt} \cdot dt = \frac{1}{2} c v^2 \Big|_{t_0}^t = \frac{1}{2} c v_{(t)}^2 - 0$$

$$W = \frac{1}{2} c V_{(t)}^2$$

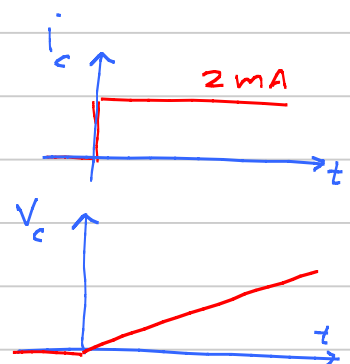
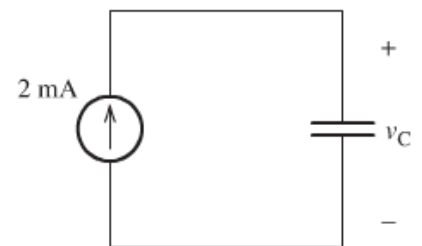
Important properties of capacitors

1. A capacitor behaves as an open circuit when its voltage is constant.
2. The voltage across a capacitor cannot change instantaneously. (Capacitor voltage is continuous).

Ex. At $t=0$, a 2-mA source is applied to an uncharged 2 micro-farad capacitor. Determine
a) the capacitor voltage at $t=30$ ms.
b) the energy stored in the capacitor at $t=30$ ms.

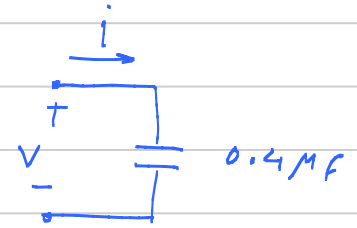
$$\begin{aligned} \text{a) } V &= \frac{1}{c} \int_0^{.03} i dt + V_0 \\ V &= \frac{1}{2 \times 10^{-6}} \int_0^{.03} .002 dt + 0 \\ &= \frac{1}{2 \times 10^{-6}} \left(.002 t \Big|_0^{.03} \right) = 30V \end{aligned}$$

$$\text{b) } W = \frac{1}{2} c V^2 = \frac{1}{2} (2 \times 10^{-6}) (30)^2 = 0.9 \text{ mJ}$$



Ex: The voltage of a 0.4 microfarad capacitor is:

$$V = \begin{cases} 25 \text{ V} , & t < 0 \\ A_1 t e^{-1500t} + A_2 e^{-1500t} , & t \geq 0 \end{cases}$$



and the current at \$t=0\$ is 90 mA.

- Find the initial energy stored in the capacitor.
- Find \$A_1\$ and \$A_2\$.

$V_c(t=0^-) = V_c(t=0^+)$, capacitor voltage is continuous.

$$\begin{aligned} \text{a) } W &= \frac{1}{2} C V_0^2 \\ &= \frac{1}{2} (0.4 \times 10^{-6}) (25)^2 = \underline{125 \mu J} \end{aligned}$$

$$\text{b) } \left(\cancel{A_1 t e^{-1500t}} + A_2 e^{-1500t} \right) \Big|_{t=0} = 25$$

$$\underline{A_2 = 25}$$

$$\begin{aligned} i &= C \frac{dV}{dt} = (0.4 \times 10^{-6}) \frac{d}{dt} \left[A_1 t e^{-1500t} + 25 e^{-1500t} \right] \\ &= 0.4 \times 10^{-6} \left[A_1 (e^{-1500t} - 1500 t e^{-1500t}) - 25 \times 1500 e^{-1500t} \right] \Big|_{t=0} = 90 \text{ mA} \end{aligned}$$

$$\underline{A_1 = 262.5 \frac{\text{KV}}{\text{s}}}$$